ALTASTEEL HYDROGEN DIFFUSION AND MICROSTRUCTURE

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Background

The CCWJ is investigating the concentration of hydrogen in AltaSteel's grinding media products. These products are close to AISI 1070 chemistry and are continuously cast into 8" x 8" billets. The billets are then hot rolled to shape, before being reheated and quenched during the finishing step. Throughout this process, hydrogen content in the steel decreases as it diffuses out. Currently, slow cooling is used as a means of ensuring hydrogen content is lowered to adequate levels by the time the product is finished.

Figure 1: Photo of AltaSteel grinding rod product.

Hydrogen content above approximately 0.8 ppm in grinding media can lead to hydrogen embrittlement. This reduces the ductility and the stress required to initiate and propagates cracks, allowing subcritical cracks to grow. Premature failures of grinding media cause production shutdowns and elevated safety risks. However, different kinds of hydrogen in the steel exist. Diffusible hydrogen exists as highly mobile H atoms, residual hydrogen is trapped at grain boundaries, dislocations, carbide interfaces, and impurities, while fixed hydrogen exists in stable hydrides. Each form of hydrogen has different diffusion activation energies.

Objectives

- AltaSteel billet microstructures will be identified and pictured along the section width of the billet.
- Diffusible and trapped hydrogen content during heating will be measured and analyzed to calculate activation energies for diffusion.
- Microstructure characterization and hydrogen content data will compliment ongoing work in the development of a hydrogen diffusion model for AltaSteel's manufacturing process.

Methodology

A billet slice provided from AltaSteel was cut into samples of approximately 1 x 0.8 cm and hot mounted. Samples were polished using a final polishing step of 0.06 µm colloidal silica and etched with 2% nital before examination with a metallurgical light microscope. Grain size was measured by counting grain boundaries along straight lines of known length randomly imposed on micrographs.

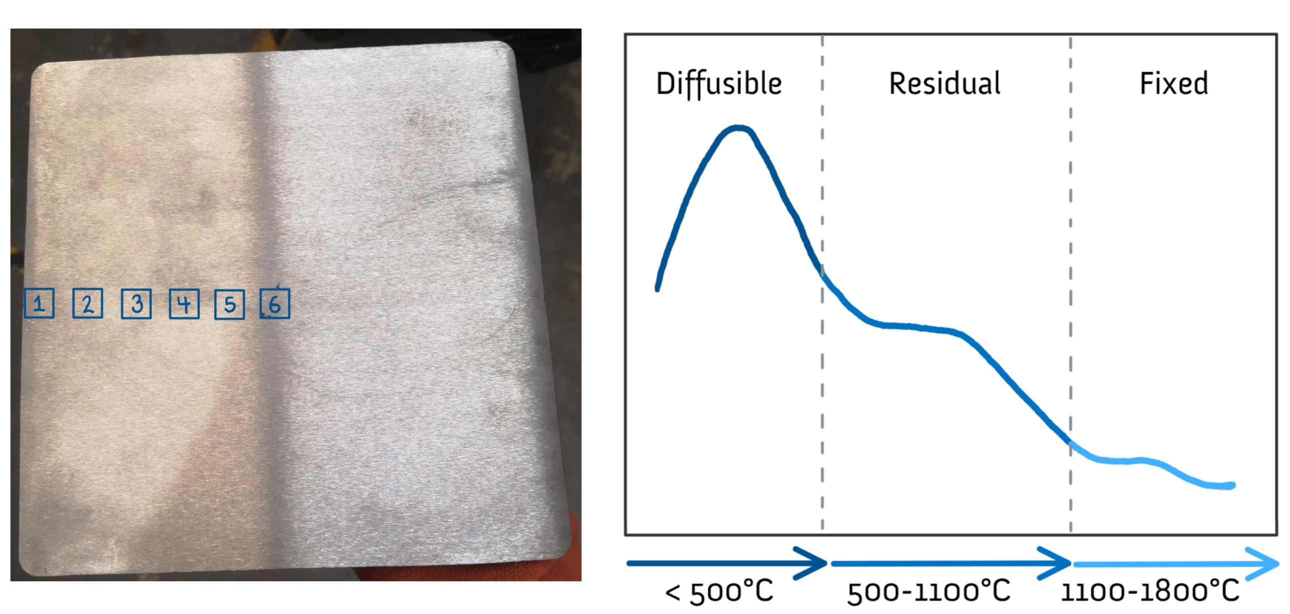

Figure 2: Billet slice with sampling locations indicated (left) and a graph depicting the hydrogen forms and temperatures they are activate at (right).

Another billet slice provided by AltaSteel was cut into strips and cleaned before testing in a Galileo G8 machine by Bruker. Hydrogen concentration of the off-gas was determined by the detector unit over the course of the heating of the steel samples.

Results

Microstructure

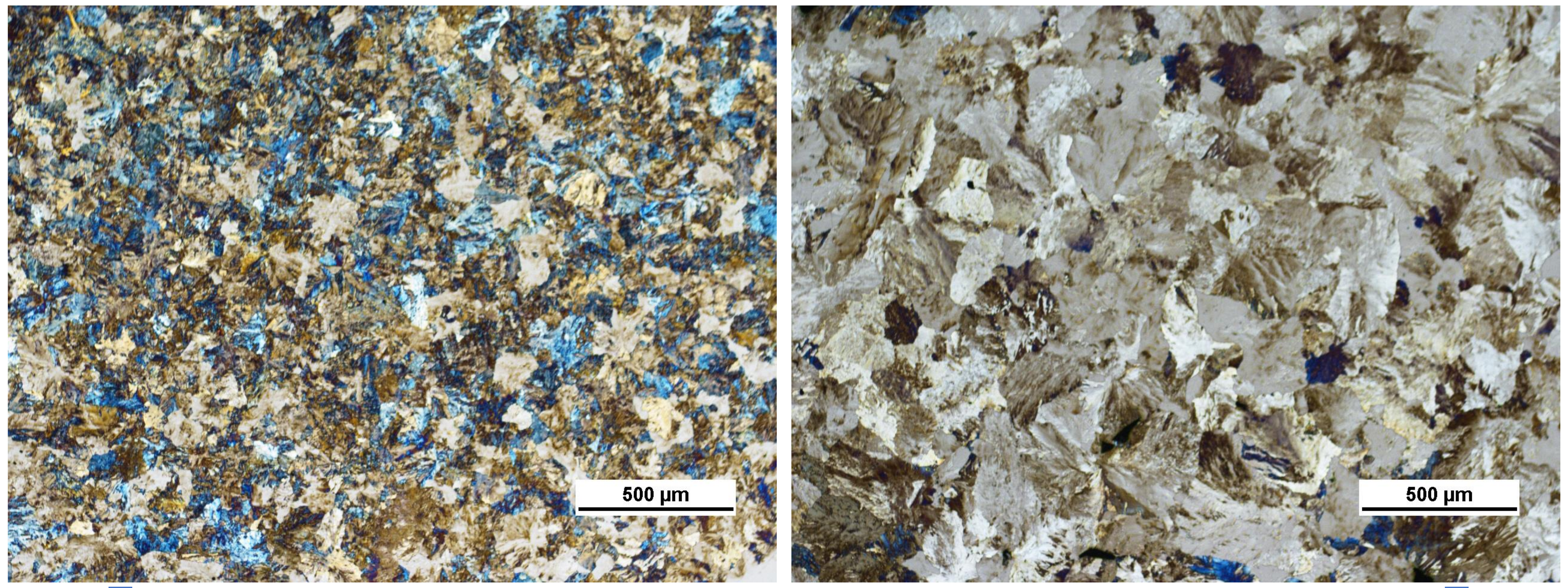

Figure 3: Micrographs showing grain size at sample locations 1 and 6. Original magnification: 5x.

Table 1: Hardness values and average grain size at the 6 sample locations along the billet width.

Sample Location:	1	2	3	4	5	6
Average Grain Size (µm):	93.3	143	184	189	206	259

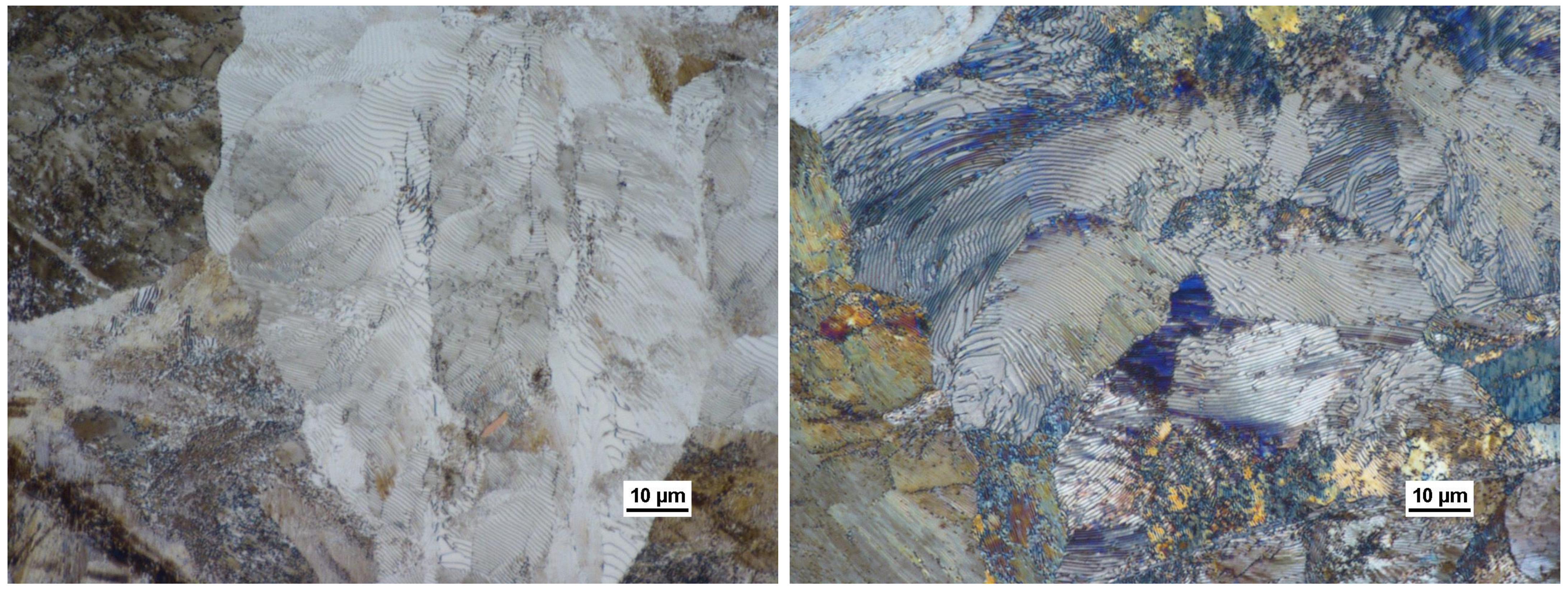

Figure 4: Micrographs showing high magnification view of grains at sample locations 1 and 6. Original magnification: 100x.

Activation Energies

$$\frac{\partial \ln(\phi/T_p^2)}{\partial(1/T_p)} = -\frac{E_a}{R}$$

Where:
Φ, Heating rate, K/s
T_p, Peak temperature, K
R, Gas constant, J/molK

Calculation:
E_a, Diffusible Hydrogen: 8.88 kJ/mol
E_a, Trapped Hydrogen: 13.1 kJ/mol

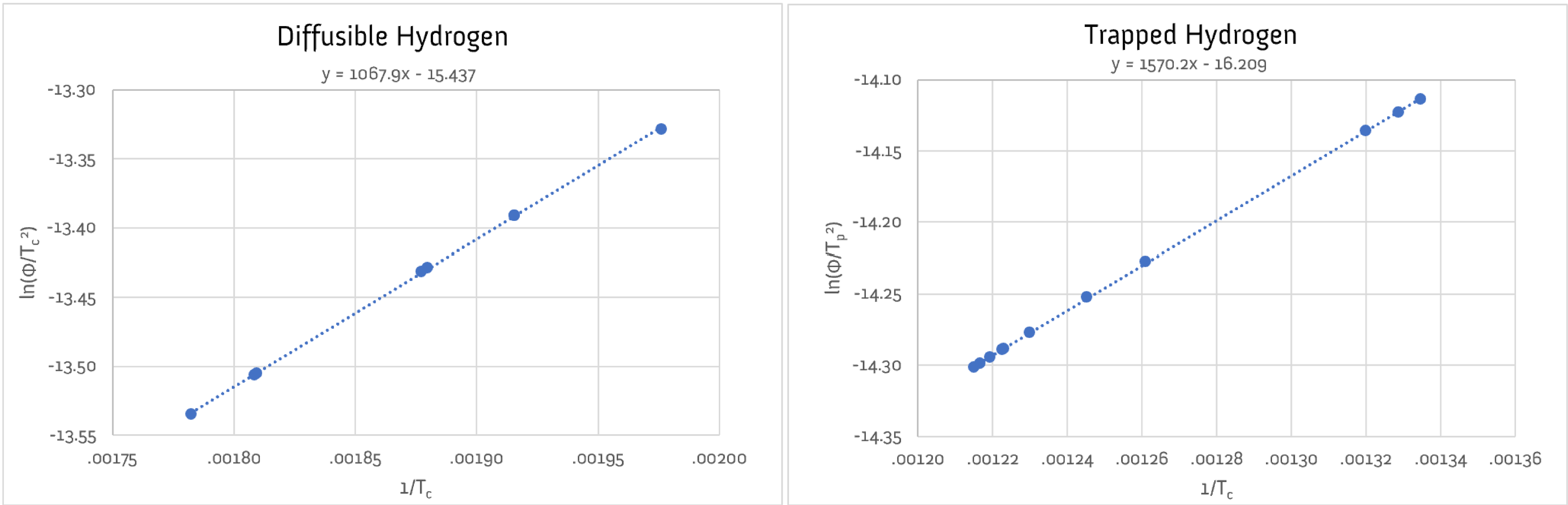

Figure 5: Plots of $\ln\left(\frac{\Phi}{T_p^2}\right)$ vs $\left(\frac{1}{T_p}\right)$ used to determine $\frac{E_a}{R}$, the slope.

Discussion

Microstructure

- A predominantly pearlitic microstructure was found.
  - Nital etch revealed wormy lines of cementite interspacing light-coloured regions of ferrite.
  - The near-eutectic composition of the steel aligns with this finding.
- The grain size was found to increase by a factor of approximately 3 between the outside and center of the billet.
  - The slower cooling rate of the inside of the billet gives the innermost grains longer time to grow, leading to grain size increasing with decreasing cooling rate from the outside to the inside of the billet.

Activation Energies

- Diffusion activation energies for diffusible and trapped (residual) hydrogen were determined.
  - Trapped hydrogen activation energy was found to be higher than diffusible hydrogen activation energy, which is in line with the prediction that trapped hydrogen requires higher temperatures and more energy to diffuse.
- Average peak temperatures at which maximum hydrogen escaping occurred were 535°C and 797°C for diffusible and trapped hydrogen, respectively.
  - Observations aligned with the prediction of separate peaks; however, the diffusible hydrogen peak occurred slightly higher than expected.
  - The diffusible hydrogen peak was observed to be smaller than the residual hydrogen peak, likely due to significant diffusion of mobile hydrogen during the casting and hot stacking steps of manufacturing.

Conclusion & Future Work

The results of this project are to be used to compliment the development of a hydrogen diffusion model for AltaSteel's manufacturing process. There are 3 main components for creating such a model:

- Hydrogen diffusion activation energies
  - Were determined for both diffusible and trapped hydrogen.
- Pre-exponential diffusion coefficients
  - Obtained through the microstructure identification and the total hydrogen measurements done with the Galileo G8 machine by Bruker.
- Time & Temperature
  - These are parameters that are determined by AltaSteel's manufacturing process

Future Work

- Scanning electron microscopy (SEM) could be used to obtain a more detailed and accurate representation of AltaSteel's billet and grinding rod microstructures.
- Energy dispersive x-ray (EDX) analysis could be used in conjunction with SEM to ascertain the chemical makeup of inclusions and regions of interest identified during optical microscopy.
- This further examination could help to establish a baseline for comparison during quality investigations and aid in improving the understanding of failure mechanisms for grinding rod product.

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